Sunada's Method

Using group theory to construct isospectral drums

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Outline

1. Introduction: What does it mean to "hear" the shape of a drum?
2. Sunada's Machine
3. Why this works
4. Pushing it further

Introduction: What does it mean to "hear" the shape of a drum?

Can you hear the shape of a drum?

- In 1963, Mark Kac asked in a now-famous paper, "Can one hear the shape of a drum?"
- What this means is: Can we recover the shape (isometry class) of a drum (planar domain, or more generally manifold) from its sound (natural frequencies of vibration, aka Laplace eigenvalues)

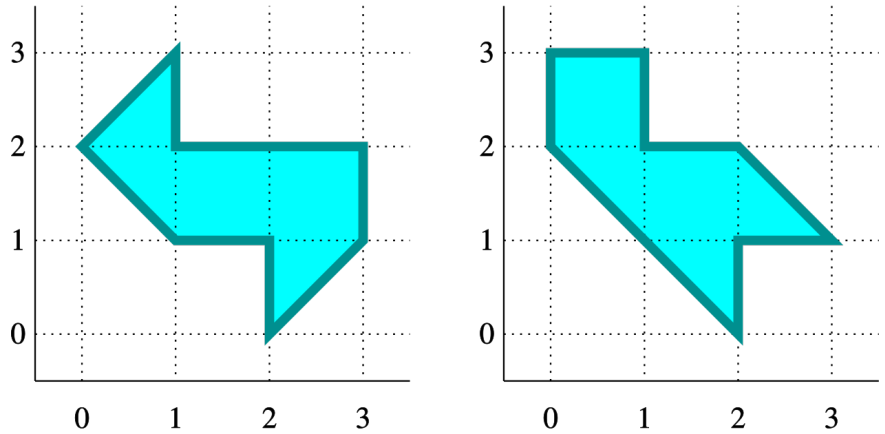


Figure: Gordon, Webb, and Wolpert's counterexample

Solving the wave equation

Why should Laplace eigenvalues correspond to natural frequencies of vibration?

- The *wave equation* in n $(= 2)$ space variables and 1 time variable is given by

$$U_{tt} = \Delta U$$

  where $\Delta := \frac{\partial}{\partial x^2} + \frac{\partial}{\partial y^2}$ is the Laplacian operator
- The solution space to this equation can be decomposed into functions that satisfy $\Delta f = \lambda f$, i.e. eigenfunction of the Laplacian. This is the Laplace spectrum
- Thus, any "vibration" (solution to the wave equation) is built up of superposing natural frequencies (Laplace eigenfunctions)

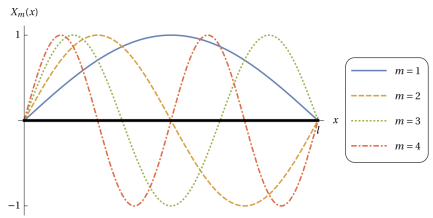


Figure: First four eigenfunctions on a string

A slight generalization

- We will think about this problem in the context of manifolds, not necessarily just planar domains or Euclidean domains
- We can define a Laplacian on any manifold that is equipped with a metric (which is a way of measuring angles, and thus a way of measuring distances)
- This is a useful generalization! The first counterexample to Kac's conjecture was found by Milnor in 1966, involving 16-dimensional tori. The planar examples were found much later and only because of the theory built in more general spaces

Just for fun

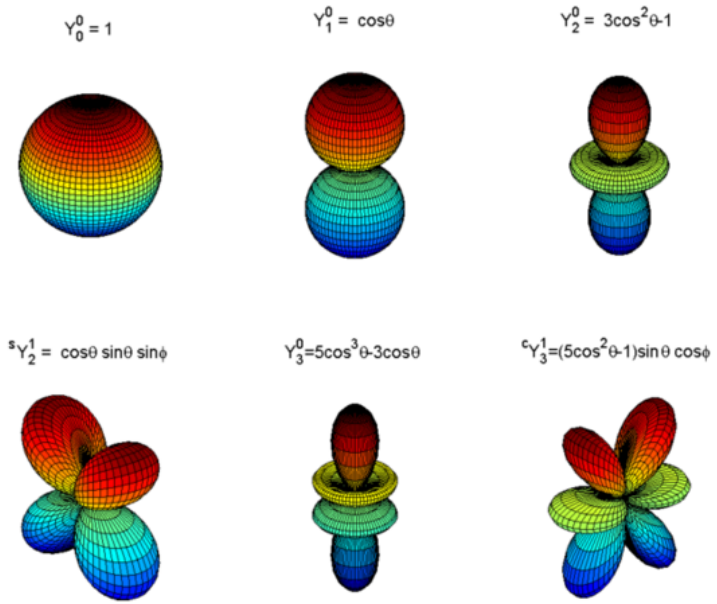


Figure: Laplace eigenfunctions on the sphere are called "spherical harmonics." Do these remind you of something?

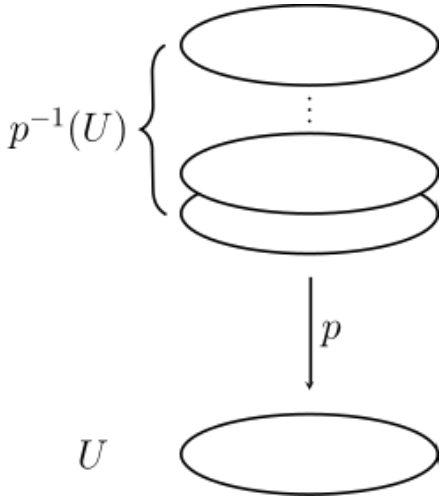
Sunada's Machine

Isospectral manifolds

- Two manifolds that have the same Laplace spectra are called isospectral
- To see the extent to which we can hear the shape of a drum, it is informative to try to be the adversary: construct non-isometric isospectral shapes
- There were a few examples of these here-and-there in the years after Milnor, but the first result that gave a systematic way of constructing them was Sunada's method

Covering spaces

- Sunada's method relies on the theory of covering spaces, in particular, the correspondence between fundamental groups and covering spaces



- Given a manifold M_0 and its universal cover M, to every subgroup H of $\pi_1(M_0)$ we can associate a cover M_H of M_0 obtained by forming the quotient space H/M

Covering spaces

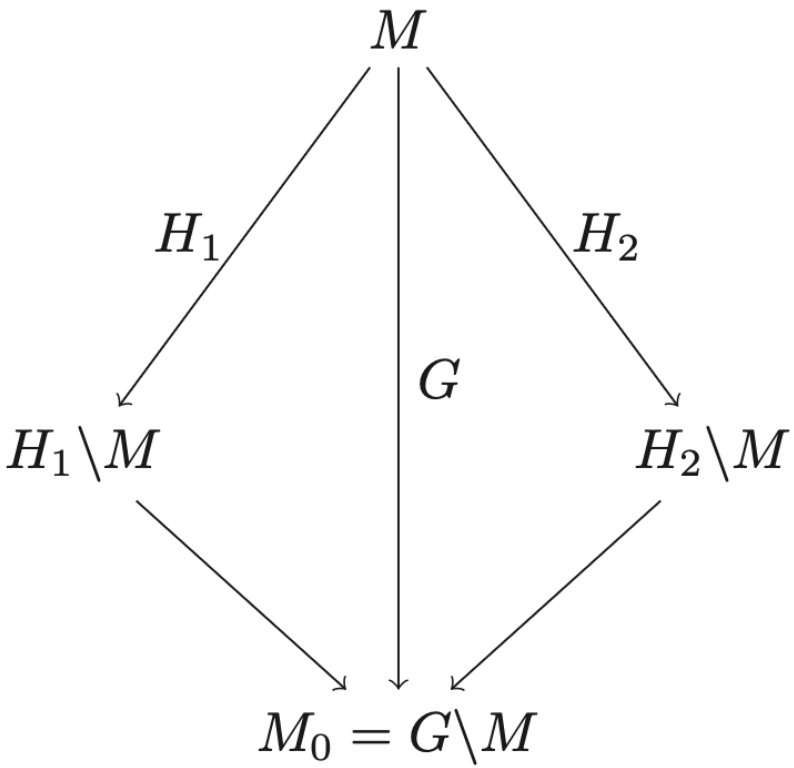


Figure: This is the setup we will be thinking about. There is a manifold M_0 with universal cover M and fundamental group G. From the data of subgroups of G, we can pass to intermediate covers of M. We will be looking at two subgroups H_1 and H_2 and comparing their corresponding intermediate covers

Almost-Conjugate subgroups

If H_1 and H_2 are conjugate, then the intermediate covers M_1 and M_2 are isometric - hence isospectral. But Sunada observed that there is a weaker condition that suffices to give iso*spectral* covers

Definition

We say that two subgroups $H_1, H_2 \leq G$ are *almost conjugate* in G if they meet each conjugacy class in G the same number of times.

Sunada's Theorem

Theorem

If M_0 is a manifold with fundamental group G and H_1 and H_2 are almost conjugate subgroups of G, then the corresponding covering spaces M_1 and M_2 are isospectral

Since this is trivially true if H_1 and H_2 are conjugate in G, it motivates the following definition

Definition

We call a triple of groups G, H_1, H_2 a *Sunada triple* if H_1 and H_2 are almost-conjugate subgroups of G that are not conjugate

(This is also called a Gassmann-Sunada triple; this technique was used in number theory using the algebraic Galois correspondence as early as 1925 and indeed that is what inspired Sunada)

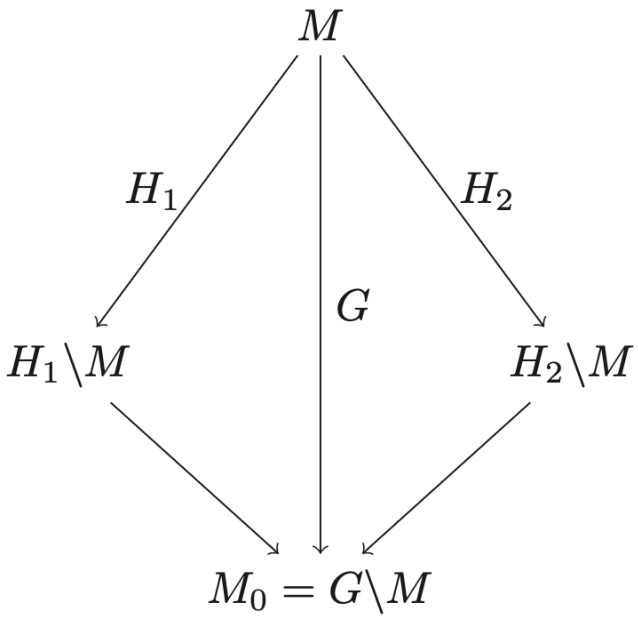
The Game

With Sunada's theorem, there is an easy game to construct isospectral manifolds

1. Find a Sunada triple G, H_1, H_2. There are many of these, see Conway, Doyle, et al.
2. Find a manifold M_0 with fundamental group G. Every finite group is realizable as a fundamental group of a manifold of dimension at most four, so this is doable
3. Let M denote the universal cover of M_0. Construct the covers H_1/M and H_2/M. They are guaranteed to be isospectral
4. It requires a little bit more trickery to prove they are not isometric, but there is a construction of putting a "nowhere homogeneous metric" on M so that no intermediate covers are "accidentally isometric"

Why this works

Upstairs and Downstairs



The idea behind all proofs of the Sunada method relies on the same idea: data "downstairs" is the same as data "upstairs" that is invariant under the group we have quotiented by. There are proofs using heat invariants, geodesics, etc. We will give the representation-theoretic proof

Eigenfunctions upstairs and downstairs

- Let $L_2(M)$ denote the space of square-integrable functions on M. The groups G acts on this via $f \mapsto (x \mapsto f(g^{-1}x))$. Then $L_2(M_i)$ is isomorphic to the subspace of $L_2(M)$ invariant under the action of H_i
- The Laplacian commutes with isometries, so $\Delta(g \cdot f) = g \cdot \Delta f$. Thus, if f is a Laplacian eigenfunction of eigenvalue λ, so is $g \cdot f$. The action of G leaves each λ-eigenspace $E_\lambda(M)$ invariant, giving us a representation of G
- A function is an eigenfunction on M_i if and only if it corresponds to an eigenfunction f of M such that $H_i \cdot f = f$
- A number λ is an eigenvalue of M_i if and only if it is an eigenvalue of M, and its multiplicity is the dimension of the subspace fixed by H_i

Representation theory

- In representation theoretic terms, the multiplicity of λ as an M_i eigenfunction is the dimension of the trivial representation of H_i in $E_\lambda(M)$
- We write this as

$$\langle \mathbf{1}_{H_i}, \mathrm{Res}^G_{H_i}(E_\lambda(M)) \rangle$$

- It turns out that $\mathrm{Res}^G_{H_i}(E_\lambda(M))$ only depends on the number of times H_i intersects each conjugacy class of G
- Thus, if H_1 and H_2 are almost conjugate, for every λ eigenvalue of Δ on M, it have the same multiplicity as an eigenvalue of M_1 and M_2, and so M_1 and M_2 are isospectral

The proof, for those who are interested

- Via Frobenius reciprocity,

$$\langle \mathbf{1}_{H_i}, \mathrm{Res}^{G}_{H_i}(E_\lambda(M)) \rangle = \langle \mathrm{Ind}^{G}_{H_i}(\mathbf{1}_{H_i}), E_\lambda(M) \rangle$$

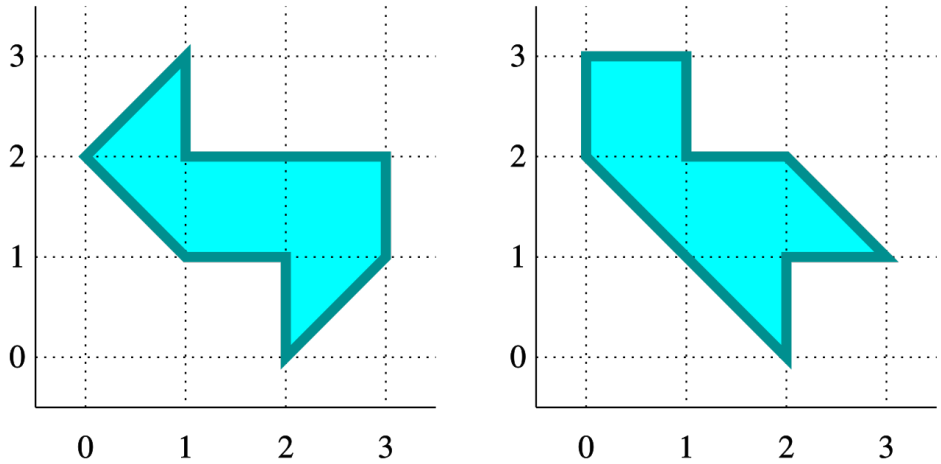
- By some character theory (see Fulton and Harris, section 3.3)

$$\chi_{\mathrm{Ind}^{G}_{H} W}(g) = \frac{[G : H]}{\#[g]} \cdot \#([g] \cap H)$$

- Thus almost conjugate subgroups are *precisely* the subgroups which induce the same representations of their overgroup

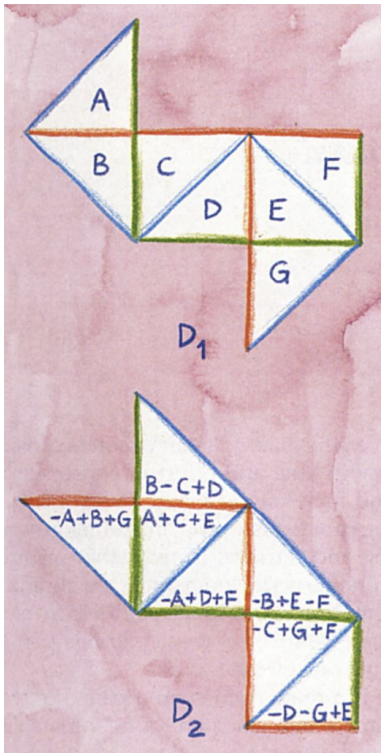
Pushing it further

But what about planar domains?



The original questions were about drums that are planar domains. These are not manifolds, they are manifolds with boundary! With corners!! Yet, this construction is what eventually led to the winning counterexample

Transplantation Maps



The way Gordon, Webb, and Wolpert constructed their drums was by using "transplantation maps" to transplant eigenfunctions from one domain to the other. The data of these transplantation maps came from the almost conjugate subgroups of Sunada's method!

We cannot hear the shape of a drum

Moral: If you want to solve a problem in two dimensions, try sixteen first

Actually....

While we have examples of isospectral non-isometric planar domains, the following questions are still open, even in 2D:

1. Can you hear the shape of a convex drum?
2. Can you hear the shape of a smooth drum?